THE DIMENSION FORMULA FOR COHEN-MACAULAY MORPHISMS OF LOCALLY NOETHERIAN SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5846-5855 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5857-5866, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The lemma that CM morphisms of finite presentation have equidimensional fibres of constant local dimension after shrinking (Morphisms, flat-finite-presentation-CM-fibres-relative-dimension) and the dimension formula for morphisms of given relative dimension (Morphisms, rel-dimension-dimension); cited inputs with the reduction written in the retained span.

Source reference. Original lemma 02NM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5866–5866. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AFE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5866–5866. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of locally Noetherian schemes which is locally of finite type and Cohen-Macaulay. For every point x in X with image y in Y,*

$$\dim_x(X) = \dim_y(Y) + \dim_x(X_y),$$

where X_y denotes the fiber over y.

Proof. After replacing X by an open neighborhood of x, there is a natural number d such that all fibers of $X \to Y$ have dimension d at every point, see Morphisms, Lemma 02NM. Then f is flat, locally of finite type and of relative dimension d. Hence the result follows from Morphisms, Lemma 0AFE. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.